

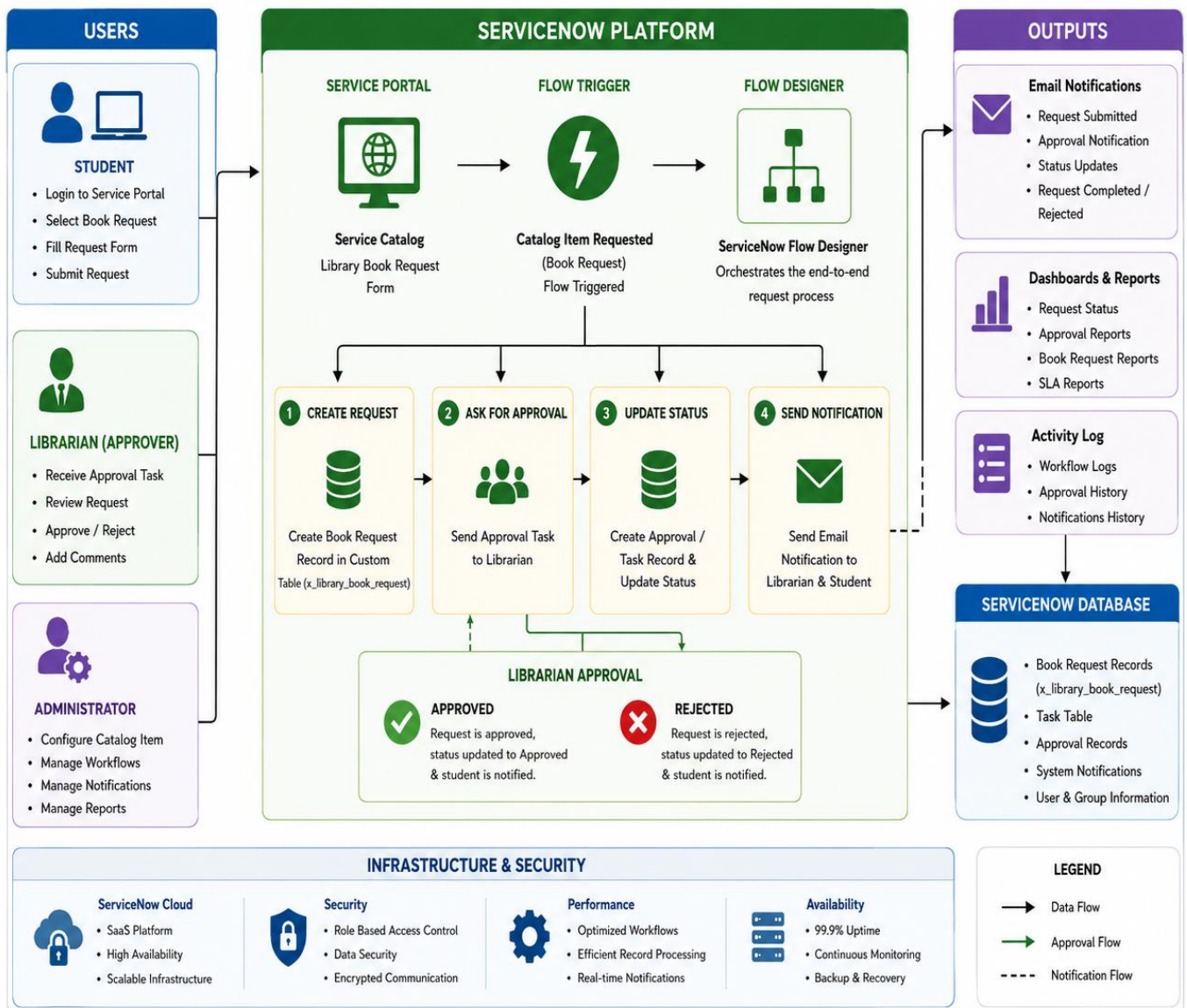
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30/06/2026
Roll no	23A51A4272
Project Name	Library Book Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

TECHNICAL ARCHITECTURE – LIBRARY BOOK REQUEST PROCESS IN SERVICENOW



Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Students submit library book requests through the Service Catalog	ServiceNow Service Catalog
2	Application Logic-1	Creates Library Book Request records automatically	ServiceNow Flow Designer
3	Application Logic-2	Validates Issue Date using Business Rule	ServiceNow Business Rule
4	Application Logic-3	Sends email notifications to the librarian after request submission	ServiceNow Notifications / Flow Designer
5	Database	Stores Library Book Request records and request details	ServiceNow Custom Table (Library Book Request)
6	Cloud Database	Cloud-based storage managed by the ServiceNow platform	ServiceNow Cloud Database
7	File Storage	Stores attachments related to book requests (if uploaded)	ServiceNow Attachment Storage
8	Infrastructure (Server / Cloud)	Application hosted on the ServiceNow cloud platform	ServiceNow Cloud Platform (SaaS)

Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source frameworks are used. The project is developed using ServiceNow's built-in platform features.	ServiceNow Platform
2	Security Implementations	Role-based access control allows only authorized students, librarians, and administrators to access the application.	ServiceNow RBAC, User Roles
3	Scalable Architecture	ServiceNow cloud architecture supports multiple library users and book requests efficiently.	ServiceNow Multi-Tenant Cloud Architecture
4	Availability	ServiceNow provides high availability and continuous access through its cloud infrastructure.	ServiceNow Cloud Services
5	Performance	Automated workflows, Business Rules, Number Maintenance, and Flow Designer reduce manual work and improve request processing speed.	ServiceNow Flow Designer, Business Rules, Number Maintenance